

Differential cryptanalysis of "An optimized novel lightweight block cipher for image encryption"

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ABSTRACT

This paper presents a security analysis of the ARX-based lightweight block cipher proposed by Mohanapriya and Nithish (*Sci Rep* 15, 36060 (2025)) for image encryption in IoT environments. The cipher employs a 64-bit key and a 64-bit block size, relying on Addition, Rotation, and XOR (ARX) operations. Our analysis demonstrates that the full-round version of this cipher is vulnerable to differential cryptanalysis. In fact, the cipher can be distinguished from a random permutation using 2^{41} chosen plaintexts. Consequently, the designers' security claims are not fully supported.

1 Introduction

A block cipher F takes a key K of fixed bit length as input and produces a bijective mapping (permutation) from an n -bit plaintext to an n -bit ciphertext, where n is the fixed bit length of the block size. Assume that k is the fixed bit length of the key size. In this case, block ciphers are described as $F_K : \{0, 1\}^n \rightarrow \{0, 1\}^n$, where $K \in \{0, 1\}^k$. The permutation F_K is expected to act completely independently for different choices of K .

A fundamental security requirement for a block cipher is that, for any given key K , the transformation F_K it defines must be computationally indistinguishable from a permutation selected uniformly at random from the set of all possible permutations on the domain. This property is characterized as the pseudo-random permutation assumption. Essentially, once a key is fixed, the cipher operates as a specific, deterministic permutation on the set of n -bit strings.

Under the condition that the key K remains secret, no adversary should possess the capability to generate a correct plaintext-ciphertext pair. In the corresponding security model, a hypothetical random permutation is selected from the set of all n -bit permutations. The adversarial objective in this framework is to determine whether the system they are interacting with contains the block cipher (with its secret, fixed key) or a genuinely random permutation, all without prior knowledge of the key.

The existence of a practical algorithm that can successfully undermine this indistinguishability property is considered a major cryptanalytic breakthrough, indicating a fundamental weakness in the cipher's design. The resistance of cryptographic algorithms against various attacks is a critical aspect that designers must examine with high precision. In addition to the structure of cryptographic algorithms, which can lead to distinguishers, designers must ensure that the designed algorithm is resistant at least to the most well-known cryptographic attacks, such as differential attack¹, linear attack², impossible differential attack^{3,4}, zero-correlation attack⁵, related-key attacks^{6,7}, and others. Clearly, any cryptographic algorithm must be designed to be resistant to these attacks; therefore, cryptographic designers must pay close attention to these basic attack models in their designs. Examples of ciphers that have been completely broken against these attacks can be found in⁸⁻¹¹.

Nowadays, the use of automated search methods, such as Mixed-Integer Linear Programming (MILP), Satisfiability Modulo Theory (SAT)-based methods, or CP-based methods, can greatly assist designers of cryptographic algorithms in improving their design¹²⁻¹⁶. These automated methods can help identify initial design errors, which is an important aspect that was not utilized in the design of the cryptographic algorithm presented in¹⁷, where security analysis is also a focus.

The development of specialized encryption algorithms for IoT¹⁸, medical imaging¹⁹, and satellite data²⁰ demonstrates a shift towards application-specific security. In line with this trend, a new block cipher has been designed in¹⁷ for image encryption purposes. The paper by Mohanapriya et al.¹⁷ proposes a lightweight, ARX-based block cipher for image encryption. The cipher uses a 64-bit master key to encrypt 64-bit plaintext blocks. Its design features a multi-stage ARX (Addition-Rotation-XOR) structure and a custom key schedule that generates five subkeys per round using bit-swapping, modular addition, and XOR operations. The encryption process consists of five rounds, each with five internal stages that perform operations on 16-bit data segments.

The designers claim the cipher achieves a maximum differential probability of approximately 2^{-32} and a maximum linear bias below 2^{-8} , suggesting strong resistance to differential and linear cryptanalysis over five rounds¹⁷.

In this paper, we analyze the algorithm's security against differential cryptanalysis. Our findings demonstrate that the proposed 5-round cipher is not secure against this attack, thereby invalidating the designers' original security analysis.

1.1 Our Contribution

In this paper, we analyze the 64-bit lightweight block cipher designed in¹⁷. The designers claimed that it is resistant to differential attacks. We demonstrate that this cipher is not secure against differential cryptanalysis and thus can be distinguished from a 64-bit random permutation.

The optimal (maximum probability) differential characteristics for 1 to 5 rounds are: 1 round with probability 1, 2 rounds with 2^{-5} , 3 rounds with 2^{-14} , 4 rounds with 2^{-29} , and five rounds with 2^{-41} , which should be compared with 2^{-64} for a random permutation.

We also empirically verified the two and three-round characteristics to validate their correctness.

It is worth noting that the search for these differential characteristics was conducted using the automated SAT approach described in²¹.

1.2 Outline

In [section 2](#), the notations and a summary of the idea behind differential distinguishers, along with the structure of the target block cipher algorithm analyzed in this paper, are described. [section 3](#) presents the results of the search for differential distinguishers and the experimental tests of some of these distinguishers. Finally, the conclusion is provided in [section 4](#).

2 Preliminaries

2.1 Notations

The following notations are used throughout this paper: the bitwise exclusive-OR (XOR) operation is denoted by $\oplus$, addition modulo 2^{16} is represented by $\boxplus$, and bitwise right rotation is indicated by $\gg$. The concatenation of two bit strings is symbolized by $\|$. In the context of differential cryptanalysis, the difference between two values X_1 and X_2 is defined as $\Delta X = X_1 \oplus X_2$, where ΔX represents the differential.

2.2 Differential Cryptanalysis

Differential cryptanalysis was introduced by Biham and Shamir in 1990¹. The initial cryptanalysis approach to attacking the DES block cipher, which was more efficient than an exhaustive key search, was described in^{22,23}. The process of differential cryptanalysis involves the exploitation of the impact of a specific difference in a cipher's pair of inputs on a difference in the cipher's pair of outputs. The pair of outputs is generated by encrypting the pair of inputs with the same key. The XOR operation can serve as an illustration of the concept of differences. Output difference is the difference between the outputs of a function, whereas input difference is the difference between the inputs. For a block cipher F utilizing a specific key $K \in \{0, 1\}^k$, let $\Delta\alpha$ and $\Delta\beta$ represent n -bit blocks. The probability of a differential $(\Delta\alpha, \Delta\beta)$ for F_K , denoted as $\Delta\alpha \rightarrow \Delta\beta$, is defined as follows.

$$\Pr_{F_K}(\Delta\alpha \rightarrow \Delta\beta) = \frac{|\{x \in \{0, 1\}^n \mid F_K(x) \oplus F_K(x \oplus \Delta\alpha) = \Delta\beta\}|}{2^n}, \quad (1)$$

where n is the block size. For a random permutation $F_\$$ that maps $\{0, 1\}^n$ to $\{0, 1\}^n$, the chance that a certain output difference will happen when the input difference is not zero is around 2^{-n} . If $\Pr_{F_K}(\Delta\alpha \rightarrow \Delta\beta)$ exceeds 2^{-n} , with a substantial quantity of plaintext pairs, we may employ the differential to distinguish the block cipher from a randomly generated function.

Determining the exact probability $\Pr_{F_K}(\Delta\alpha \rightarrow \Delta\beta)$ is typically computationally prohibitive, since it would necessitate an exhaustive search over the entire plaintext space. However, by exploiting the internal construction of a block cipher and analyzing how differences propagate across its rounds, a reliable estimate of this differential probability can be derived far more efficiently.

The differential passes through the XOR and rotation operations with probability 1, but it need not pass through the modular addition operation with probability 1. The following two theorems clarify the necessary and sufficient conditions for the differential to propagate through the modular addition operation, along with the corresponding probability.

Theorem 1 (²⁴) *Let $z = x \boxplus y \pmod{2^n}$, and let $\Delta\alpha = (\Delta\alpha_{n-1}, \dots, \Delta\alpha_0)$ and $\Delta\beta = (\Delta\beta_{n-1}, \dots, \Delta\beta_0)$ be the input differences, and $\Delta\gamma = (\Delta\gamma_{n-1}, \dots, \Delta\gamma_0)$ be the output difference. Then, the probability of the differential $(\Delta\alpha, \Delta\beta) \rightarrow \Delta\gamma$ is non-zero if and only if:*

- (1) $\Delta\alpha_0 \oplus \Delta\beta_0 \oplus \Delta\gamma_0 = 0$,
- (2) for each i from 1 to $n-1$: if $\Delta\alpha_{i-1} = \Delta\beta_{i-1} = \Delta\gamma_{i-1} = b$, then $\Delta\alpha_i \oplus \Delta\beta_i \oplus \Delta\gamma_i = b$.

Theorem 2 ⁽²⁴⁾ *If the differential $(\Delta\alpha, \Delta\beta) \rightarrow \Delta\gamma$ has non-zero probability, then its probability is given by:*

$$\Pr[(\Delta\alpha, \Delta\beta) \rightarrow \Delta\gamma] = 2^{-\sum_{i=0}^{n-2} (1 - \text{eq}(\Delta\alpha_i, \Delta\beta_i, \Delta\gamma_i))},$$

where the function $\text{eq}(a, b, c)$ equals 1 if $a = b = c$, and 0 otherwise.

It is essential to observe that the computation $\Pr_{F_K}(\Delta\alpha \rightarrow \Delta\beta)$ typically omits the impact of cryptographic keys. Nevertheless, the actual probability of a specific differential depends significantly on the secret key selection, making it crucial to examine its statistical distribution. While one potential approach is to calculate differential probabilities across the entire key space, such computation remains computationally prohibitive in practice. This limitation motivates adopting the expected differential probability.

For a block cipher F_K that uses a k -bit key, the expected differential probability (EDP) of an r -round differential $(\Delta\alpha, \Delta\beta)$ is defined as:

$$\text{EDP}(\Delta\alpha \rightarrow \Delta\beta) = 2^{-k} \sum_{K \in \{0,1\}^k} \Pr_{F_K}(\Delta\alpha \rightarrow \Delta\beta)$$

Security proofs against differential cryptanalysis frequently rely on the Hypothesis of Stochastic Equivalence²⁵, which posits that for most keys K , the actual differential probability of any characteristic closely approximates its EDP. However, practical evidence suggests that this assumption may not be universally applicable^{15,26}.

In subsection 3.2, we conduct experimental validation of 2- and 3-round differential characteristics for the target cipher algorithm. By testing with 512 randomly selected keys and varying sizes of random data pairs, we empirically verify the accuracy of the derived differential characteristics.

A brief description of the lightweight block cipher analyzed in this paper is provided in the following section.

2.3 A brief description of the algorithm proposed in¹⁷

The target algorithm is an ARX (Addition–Rotation–XOR) block cipher designed for image encryption in constrained hardware. It operates on 64-bit plaintext blocks under a 64-bit master key. It employs a multi-stage intra-round structure and a recursive subkey expansion to achieve rapid diffusion while remaining implementation-efficient. The designers present a five-round iterative network; each round consists of five sequential stages, and each stage consumes a distinct 16-bit subkey, producing a fine-grained per-stage key dependency intended to accelerate diffusion and avoid a simple, repetitive structure across rounds. The following stepwise description summarizes the per-stage/per-round encryption layout (words are 16 bits, and additions are modulo 2^{16}) as implemented in¹⁷.

1. **State partitioning.** Represent the 64-bit round input as four 16-bit words $P_{11}, P_{12}, P_{13}, P_{14}$.
2. **Stage 1 transformation (produces M_{11} – M_{14}).**

$$(a) \ M_{11} \leftarrow (P_{11} \ggg 7) \boxplus k_1^{(i)}$$

$$(b) \ M_{12} \leftarrow (P_{12} \ggg 2) \oplus M_{11}$$

$$(c) \ M_{13} \leftarrow (P_{13} \ggg 7) \oplus M_{12}.$$

$$(d) \ M_{14} \leftarrow (P_{14} \ggg 2) \oplus M_{13}.$$

- (e) Reorder the quarter outputs to form the 64-bit intermediate outputs $(M_{12} \parallel M_{13} \parallel M_{14} \parallel M_{11})$ and forward it to the next stage.

Figure 1 illustrates the details of Stage 1 where values M_{11} – M_{14} are derived from the main text.

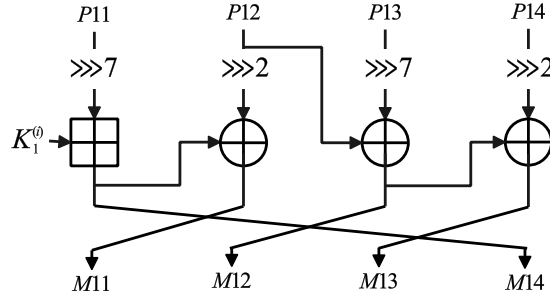


Figure 1. Stage 1 transformation (produces M_{11} – M_{14}).

3. **Stages 2–5.** Repeat the same sequence of operations as in Stage 1 for stages $s = 2$ to 5, using subkeys $k_2^{(i)}, \dots, k_5^{(i)}$ respectively; the output of stage s becomes the input to stage $s + 1$. After stage 5, the output of the round is passed to the next round (or becomes the ciphertext after round 5). Decryption involves applying inverse rotations, modular subtraction (instead of addition), and reversing the stage subkeys.

The key schedule expands the 64-bit master key into five 16-bit stage subkeys per round via a compact permutation, addition, and XOR pipeline. The published algorithm implements this expansion iteratively, so that the subkeys generated for round i feed into the key material for round $i + 1$. The principal steps of the schedule are:

1. **Initial decomposition.** Label the 64-bit master key from the LSB as sixteen 4-bit nibbles $R_1, \dots, R_{16}$.
2. **Form four 16-bit intermediates.** Concatenate the nibbles into four 16-bit words using the predefined mapping

$$C_1 = (R_1 \parallel R_5 \parallel R_9 \parallel R_{13}), C_2 = (R_3 \parallel R_7 \parallel R_{11} \parallel R_{15}), C_3 = (R_2 \parallel R_6 \parallel R_{10} \parallel R_{14}), C_4 = (R_4 \parallel R_8 \parallel R_{12} \parallel R_{16}).$$
3. **Bit-swap permutations.** Apply a distinct bitswap (permutation) to each C_j to obtain B_1, B_2, B_3, B_4 , thereby dispersing and shuffling key bits before arithmetic mixing. Bits for C_j permute as follows:

$$\begin{aligned}
 C_1 : [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15] &\rightarrow B_1 : [15, 6, 14, 5, 13, 4, 12, 3, 11, 2, 10, 1, 9, 0, 8, 7] \\
 C_2 : [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15] &\rightarrow B_2 : [8, 9, 10, 11, 4, 5, 6, 7, 12, 13, 14, 15, 0, 1, 2, 3] \\
 C_3 : [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15] &\rightarrow B_3 : [7, 6, 15, 5, 14, 4, 13, 3, 12, 2, 11, 1, 10, 0, 9, 8] \\
 C_4 : [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15] &\rightarrow B_4 : [3, 2, 1, 0, 7, 6, 5, 4, 11, 10, 9, 8, 15, 14, 13, 12].
 \end{aligned}$$

4. **Arithmetic/XOR mixing (subkey generation).** Compute intermediate subkeys

$$\begin{aligned}
 k_1^{(i)} &\leftarrow (B_1 \boxplus B_2) \boxplus B_3, \\
 k_2^{(i)} &\leftarrow B_3 \oplus B_2, \\
 k_3^{(i)} &\leftarrow B_1 \oplus B_3, \\
 k_4^{(i)} &\leftarrow B_1 \boxplus B_4,
 \end{aligned}$$

and then form the fifth stage subkey as $k_5^{(i)} \leftarrow k_1^{(i)} \oplus k_2^{(i)} \oplus k_3^{(i)} \oplus k_4^{(i)}$.

5. **Inter-round feedback.** Concatenate $k_1^{(i)} \parallel k_2^{(i)} \parallel k_3^{(i)} \parallel k_4^{(i)}$ to form the 64-bit input key for the next round's schedule: $\text{key_in}^{(i+1)}$. Iterating this process provides inter-round key dependence and reduces the risk of periodic repetition across rounds.

The published evaluation reports rapid difference propagation and strong statistical properties: empirical differential experiments (encrypting 10^5 plaintext pairs with fixed input difference) show a high $\text{DP}_{\max}$ in the first round (≈ 0.5476) that falls precipitously to the order of 10^{-5} by the second round and remains at that order for deeper rounds, and the authors conclude that five rounds are sufficient for the intended image-encryption application given the measured differential and linear biases.

3 Automated Search for Differential Characteristics of ¹⁷

We use the SAT model proposed in ²¹ to find differential characteristics of the cipher. While that paper provides a complete description of modeling the algorithm's differential behavior, here we present only a brief explanation of modeling modular addition and the weight constraint.

Modeling Modular Addition

Theorem 1 shows the possible input and output differences for modular addition. Specifically, when the input differences are $\Delta\alpha = (\Delta\alpha_{n-1}, \dots, \Delta\alpha_0)$ and $\Delta\beta = (\Delta\beta_{n-1}, \dots, \Delta\beta_0)$, and the output difference is $\Delta\gamma = (\Delta\gamma_{n-1}, \dots, \Delta\gamma_0)$. Two conditions must be satisfied. These two conditions can be written as the following equations.

For condition (1):

$$\begin{aligned}\overline{\Delta\alpha_0} \vee \Delta\beta_0 \vee \Delta\gamma_0 &= 1 \\ \Delta\alpha_0 \vee \overline{\Delta\beta_0} \vee \Delta\gamma_0 &= 1 \\ \Delta\alpha_0 \vee \Delta\beta_0 \vee \overline{\Delta\gamma_0} &= 1 \\ \overline{\Delta\alpha_0} \vee \overline{\Delta\beta_0} \vee \overline{\Delta\gamma_0} &= 1.\end{aligned}$$

For condition (2):

$$\left. \begin{aligned}\Delta\alpha_i \vee \Delta\beta_i \vee \Delta\gamma_i \vee \Delta\alpha_{i+1} \vee \Delta\beta_{i+1} \vee \Delta\gamma_{i+1} &= 1 \\ \Delta\alpha_i \vee \overline{\Delta\beta_i} \vee \Delta\gamma_i \vee \Delta\alpha_{i+1} \vee \Delta\beta_{i+1} \vee \Delta\gamma_{i+1} &= 1 \\ \overline{\Delta\alpha_i} \vee \Delta\beta_i \vee \Delta\gamma_i \vee \Delta\alpha_{i+1} \vee \Delta\beta_{i+1} \vee \Delta\gamma_{i+1} &= 1 \\ \overline{\Delta\alpha_i} \vee \overline{\Delta\beta_i} \vee \overline{\Delta\gamma_i} \vee \Delta\alpha_{i+1} \vee \Delta\beta_{i+1} \vee \Delta\gamma_{i+1} &= 1 \\ \Delta\alpha_i \vee \Delta\beta_i \vee \Delta\gamma_i \vee \overline{\Delta\alpha_{i+1}} \vee \overline{\Delta\beta_{i+1}} \vee \overline{\Delta\gamma_{i+1}} &= 1 \\ \Delta\alpha_i \vee \overline{\Delta\beta_i} \vee \overline{\Delta\gamma_i} \vee \overline{\Delta\alpha_{i+1}} \vee \overline{\Delta\beta_{i+1}} \vee \overline{\Delta\gamma_{i+1}} &= 1 \\ \overline{\Delta\alpha_i} \vee \Delta\beta_i \vee \overline{\Delta\gamma_i} \vee \overline{\Delta\alpha_{i+1}} \vee \overline{\Delta\beta_{i+1}} \vee \overline{\Delta\gamma_{i+1}} &= 1 \\ \overline{\Delta\alpha_i} \vee \overline{\Delta\beta_i} \vee \Delta\gamma_i \vee \overline{\Delta\alpha_{i+1}} \vee \overline{\Delta\beta_{i+1}} \vee \overline{\Delta\gamma_{i+1}} &= 1\end{aligned}\right\} 0 \leq i \leq n-2.$$

Theorem 2 deals with how to calculate the probability for given input and output differences. To assign the probability value according to **Theorem 2**, the following conditions must be satisfied in the SAT problem:

$$\left. \begin{aligned}\overline{\Delta\alpha_i} \vee \Delta\gamma_i \vee w_i &= 1 \\ \Delta\beta_i \vee \overline{\Delta\gamma_i} \vee w_i &= 1 \\ \Delta\alpha_i \vee \overline{\Delta\beta_i} \vee w_i &= 1 \\ \Delta\alpha_i \vee \Delta\beta_i \vee \Delta\gamma_i \vee \overline{w_i} &= 1 \\ \overline{\Delta\alpha_i} \vee \overline{\Delta\beta_i} \vee \overline{\Delta\gamma_i} \vee \overline{w_i} &= 1\end{aligned}\right\} 0 \leq i \leq n-2,$$

where w_i is $1 - eq(\Delta\alpha_i, \Delta\beta_i, \Delta\gamma_i)$ in **Theorem 2**.

Sequential encoding method

To calculate the upper bound on the total probability, the sum of the weights for each addition in rounds must be computed. For this purpose, the general sum $\sum_{j=0}^{n-1} x_j \leq k$, where x_j is a boolean variable and k is a non-negative integer, needs to be formulated

as a SAT problem in the following way²¹:

$$\left. \begin{array}{l} \overline{x_0} \vee s_{0,0} = 1 \\ \overline{s_{0,j}} = 1, \quad 1 \leq j \leq k-1 \\ \\ \overline{x_i} \vee s_{i,0} = 1 \\ \overline{s_{i-1,0}} \vee s_{i,0} = 1 \\ \left. \begin{array}{l} \overline{x_i} \vee \overline{s_{i-1,j-1}} \vee s_{i,j} = 1 \\ \overline{s_{i-1,j}} \vee s_{i,j} = 1 \end{array} \right\} 1 \leq j \leq k-1 \\ \overline{x_i} \vee s_{i-1,k-1} = 1 \\ \overline{x_{n-1}} \vee s_{n-2,k-1} = 1 \end{array} \right\} 1 \leq i \leq n-2,$$

where $s_{i,j}$, ($0 \leq i \leq n-2, 0 \leq j \leq k-1$) are auxiliary variables.

3.1 Results

In¹⁷, the designers of the proposed block cipher provided a security analysis, arguing that the cipher is resistant to differential and linear cryptanalysis. Nevertheless, our results show the weakness of this cipher against differential cryptanalysis. Based on the explanations provided in subsection 2.2 for differential attacks, if a differential characteristic with a probability greater than 2^{-n} (where $n = 64$ is the block size of the cipher) is found, then this differential characteristic can be used to distinguish the cipher from a random permutation.

In modeling the differential behavior of the target cipher in¹⁷, we fix all differentials corresponding to key bits to 0 (i.e., $(\Delta K = 0)$), construct CNF files for different numbers of stages, and solve the resulting CNF instances using the MALLOB SAT solver²⁷.

Some differential characteristics with probabilities higher than 2^{-64} for 1 to 5 rounds of the target cipher are presented in Table 1.

Table 1. Differential characteristics for 1 to 5 rounds

Round	Input Difference ($\Delta\alpha$)	Output Difference ($\Delta\beta$)	Probability
1	0x0000000000000002	0x8140050A250A8000	1
2	0x1008408040220008	0x8450912231628000	2^{-5}
3	0x0000100512852834	0x955AD5AAFDE8800A	2^{-14}
4	0x0000100512852834	0x85529616B6168400	2^{-29}
5	0x0145281A2D140000	0x8450912231628000	2^{-41}

As evident from Table 1, a differential distinguisher can be constructed for all rounds of the algorithm presented in¹⁷, indicating that this cipher can be distinguished from a random permutation, which contradicts the designers' claims.

The details of the 5-round differential characteristic, which describes the propagation of input differences through the encryption rounds, are presented in Table 5. Similarly, the corresponding characteristics (as well as the probabilities at intermediate stages for different round counts) for rounds 2, 3, and 4 are provided in Table 2, Table 3, and Table 4, respectively.

As mentioned in subsection 2.3, each round comprises five stages. The input to each stage consists of four 16-bit states. Consequently, the rows of the result tables illustrate the input and output differences at each stage, showing how they propagate through the stages. The last column indicates the log probability associated with the differential characteristic.

The details of the 5-round differential characteristic reported in Table 5 can be observed in Figure 2, Figure 3, and Figure 4. Specifically, Figure 2 illustrates the first and second rounds (Round 1 (Figure 2a) and Round 2 (Figure 2b)), Figure 3 depicts the third and fourth rounds (Round 3 (Figure 3a) and Round 4 (Figure 3b)), and Figure 4 presents the fifth round of this 5-round differential characteristic.

The following section presents experimental implementation results to verify the validity of these differential characteristics.

3.2 Experimental Validation of Differential Probabilities

To empirically estimate the expected differential transition probability $EDP(\Delta\alpha \rightarrow \Delta\beta)$, we performed the following experimental procedure.

Since the key space of the target algorithm is 2^{64} bits, we computed the EDP for 2-round and 3-round distinguishers (as shown in Table 1) using 2^9 randomly generated keys.

For each random secret key K :

Table 2. Differential characteristic for two rounds with a probability of 2^{-5}

Rounds	Stages	Differentials	Pr
1	1	$\Delta\alpha = 0x1008408040220008$	2^{-2}
	2	$0x0000040004021020$	1
	3	$0x0100000804000000$	2^{-1}
	4	$0x0000000000000002$	1
	5	$0x0000000080000000$	1
2	1	$0x0000010001000000$	1
	2	$0x0040010201020000$	1
	3	$0x0040050005008000$	1
	4	$0x8140050A250A8000$	2^{-2}
	5	$0x0040114031408102$	1
		$\Delta\beta = 0x8450912231628000$	1

Table 4. Differential characteristic for four rounds with a probability of 2^{-29}

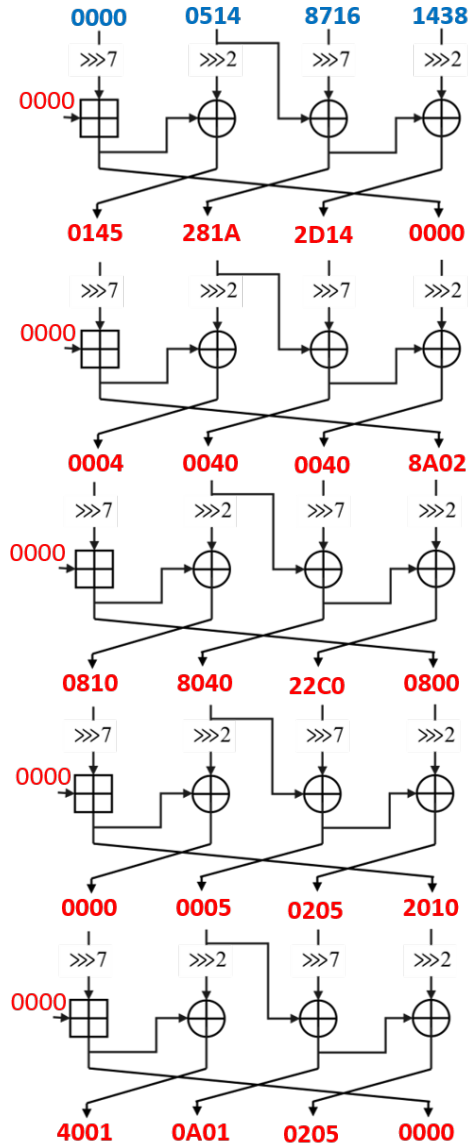
Rounds	Stages	Differentials	Pr
1	1	$\Delta\alpha = 0x0000100512852834$	1
	2	$0x44011A20102D0000$	2^{-3}
	3	$0x0400400040000288$	2^{-1}
	4	$0x1008408040220008$	2^{-2}
	5	$0x0000040004021020$	1
2	1	$0x0100000804000000$	2^{-1}
	2	$0x0000000000000002$	1
	3	$0x0000000080000000$	1
	4	$0x0000010001000000$	1
	5	$0x0040010201020000$	1
3	1	$0x0040050005008000$	1
	2	$0x8140050A250A8000$	2^{-2}
	3	$0x0040114031408102$	1
	4	$0x8450912231628000$	2^{-5}
	5	$0x034055407540A708$	2^{-2}
4	1	$0x9552D5AAF688002$	2^{-6}
	2	$0x10400452A452A52A$	2^{-1}
	3	$0x0134A11A08508020$	2^{-5}
	4	$0x0040010A2102A806$	1
	5	$0x00420548AF498000$	2^{-1}
		$\Delta\beta = 0x85529616B6168400$	1

Table 3. Differential characteristic for three rounds with a probability of 2^{-14}

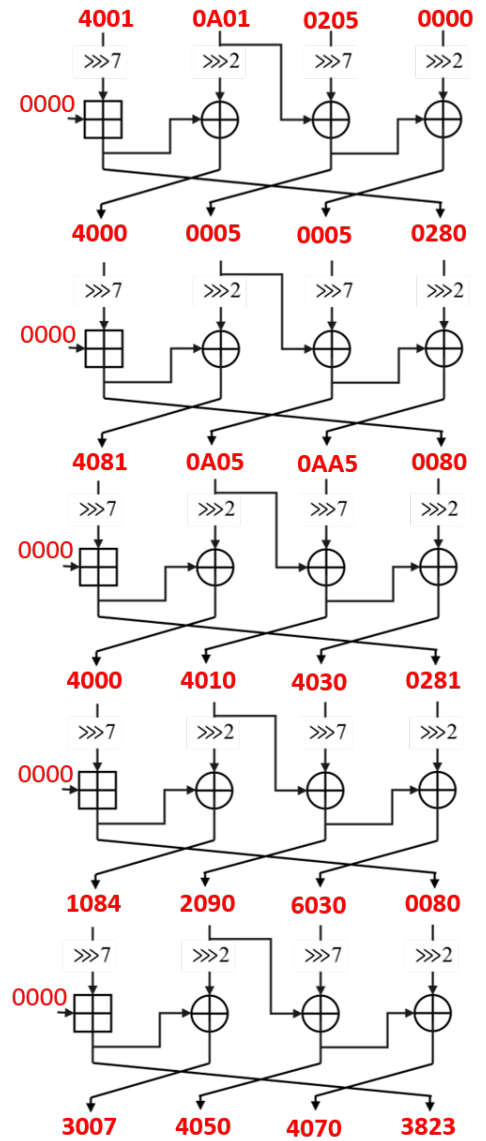
Rounds	Stages	Differentials	Pr
1	1	$\Delta\alpha = 0x0000100512852834$	1
	2	$0x44011A20102D0000$	2^{-3}
	3	$0x0400400040000288$	2^{-1}
	4	$0x1008408040220008$	2^{-2}
	5	$0x0000040004021020$	1
2	1	$0x0100000804000000$	2^{-1}
	2	$0x0000000000000002$	1
	3	$0x0000000080000000$	1
	4	$0x0000010001000000$	1
	5	$0x0040010201020000$	1
3	1	$0x0040050005008000$	1
	2	$0x8140050A250A8000$	2^{-2}
	3	$0x0040114031408102$	1
	4	$0x8450912231628000$	2^{-3}
	5	$0x054055407540A108$	2^{-2}
		$\Delta\beta = 0x955AD5AAFDE8800A$	1

Table 5. Differential characteristic for five rounds with a probability of 2^{-41}

Rounds	Stages	Differentials	Pr
1	1	$\Delta\alpha = 0x0000051487161438$	1
	2	$0x0145281A2D140000$	2^{-3}
	3	$0x0004004000408A02$	2^{-1}
	4	$0x0810804022C00800$	2^{-2}
	5	$0x0000000502052010$	1
2	1	$0x40010A0102050000$	2^{-2}
	2	$0x4000000500050280$	2^{-1}
	3	$0x40810A050AA50080$	2^{-3}
	4	$0x4000401040300281$	2^{-1}
	5	$0x1084209060300080$	2^{-6}
3	1	$0x3007405040703823$	2^{-6}
	2	$0x0A34A0D06ED81A20$	2^{-6}
	3	$0x0000100D16852834$	1
	4	$0x44031A20102D0000$	2^{-4}
	5	$0x0400400040000288$	2^{-1}
4	1	$0x1008408040220008$	2^{-2}
	2	$0x0000040004021020$	1
	3	$0x0100000804000000$	2^{-1}
	4	$0x0000000000000002$	1
	5	$0x0000000080000000$	1
5	1	$0x0000010001000000$	1
	2	$0x0040010201020000$	1
	3	$0x0040050005008000$	1
	4	$0x8140050A250A8000$	2^{-2}
	5	$0x0040114031408102$	1
		$\Delta\beta = 0x8450912231628000$	1

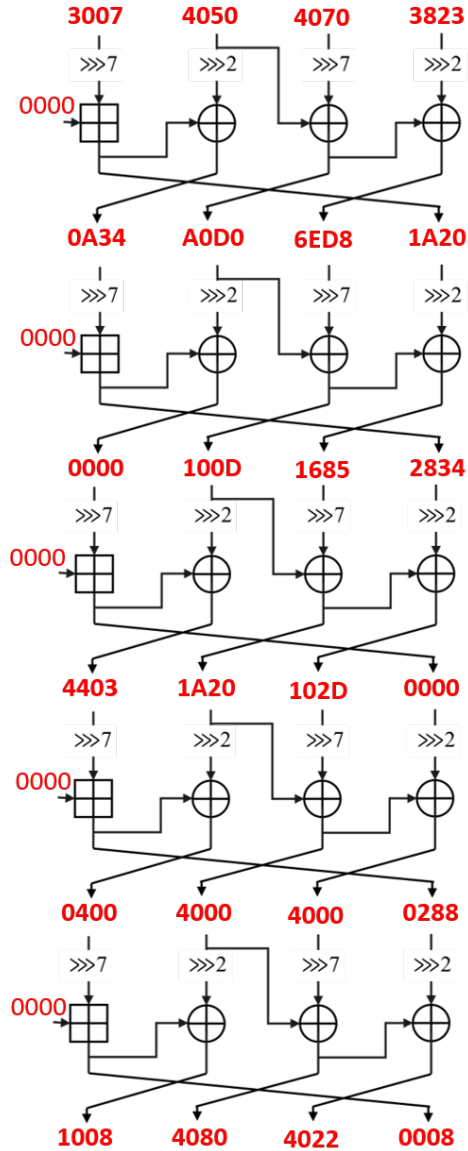


(a) First round of the 5-round differential characteristic reported in Table 5.

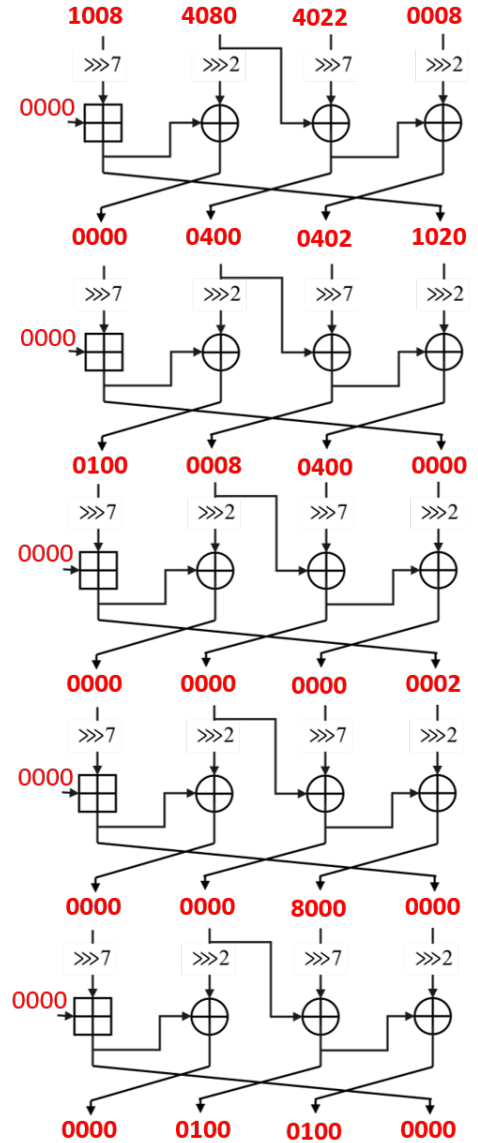


(b) Second round of the 5-round differential characteristic reported in Table 5.

Figure 2. Details of the first two rounds of the 5-round differential characteristic presented in Table 5.



(a) Third round of the 5-round differential characteristic reported in Table 5.



(b) Fourth round of the 5-round differential characteristic reported in Table 1.

Figure 3. Details of the third and fourth rounds of the 5-round differential characteristic presented in Table 5.

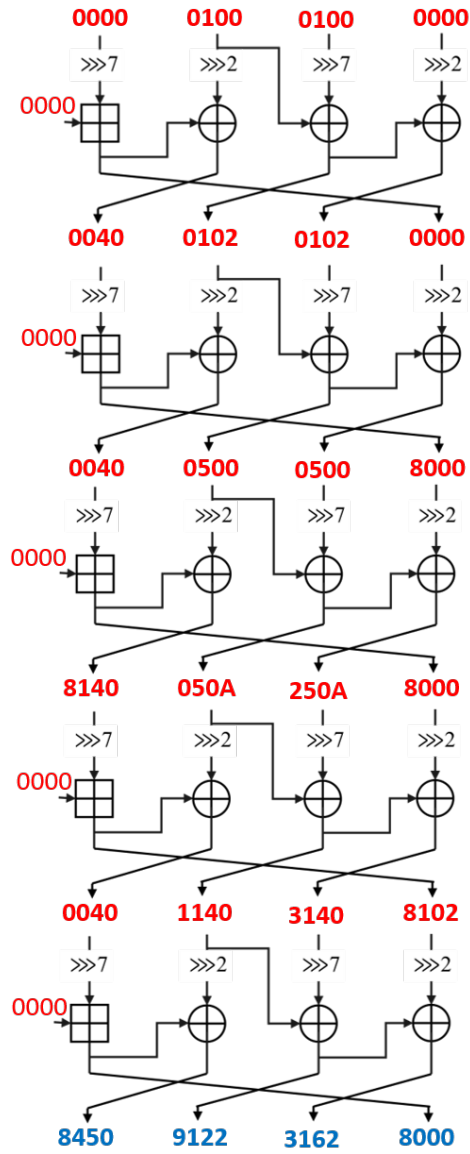


Figure 4. Fifth round of the 5-round differential characteristic reported in [Table 5](#).

- Query N random plaintexts p_i to the encryption and obtain the corresponding ciphertexts c_i .
- Query $p'_i = p_i \oplus \Delta\alpha$ to the encryption and obtain the corresponding ciphertexts c'_i .
- Count the number of ciphertext pairs (c_i, c'_i) that satisfy the output difference $c'_i \oplus c_i = \Delta\beta$.

This process was repeated over 2^9 independently generated random keys. Note that when examining an event with probability 2^{-x} using 2^x trials, the expected number of successes is 1, but this does not guarantee success with probability 1. The success probability of the distinguishing attack can be increased by allocating more resources to data and time complexity.

The experimental results for the 2-round and 3-round distinguishers are presented in Table 6. As observed in the table:

For two rounds, we tested 2^9 randomly generated keys and, for each fixed key, examined data pairs ranging from $N = 2^5$ to $N = 2^{14}$. The empirical results closely match the theoretical expectations, confirming the validity of the distinguishers proposed in this paper. Similar results hold for three rounds, as shown in Table 6, where we tested data pairs ranging from $N = 2^{14}$ to $N = 2^{25}$.

Table 6. Empirical differential probabilities $EDP(\Delta\alpha \rightarrow \Delta\beta)$ for 2-round and 3-round differential distinguishers with theoretical probabilities 2^{-5} and 2^{-14} , respectively.

Rounds	# Tested Plaintext Pairs N	Expected Matches (Theoretical)	Empirical Matches (Average over 2^9 keys)
2	2^5	2^0	$2^{-0.17}$
	2^6	2^1	$2^{1.06}$
	2^7	2^2	$2^{2.02}$
	2^8	2^3	$2^{2.81}$
	2^9	2^4	$2^{3.91}$
	2^{10}	2^5	$2^{4.87}$
	2^{11}	2^6	$2^{5.98}$
	2^{12}	2^7	$2^{6.94}$
	2^{13}	2^8	$2^{7.78}$
	2^{14}	2^9	$2^{8.99}$
3	2^{14}	2^0	$2^{-0.3}$
	2^{15}	2^1	$2^{1.13}$
	2^{16}	2^2	$2^{1.48}$
	2^{17}	2^3	$2^{2.41}$
	2^{18}	2^4	$2^{4.19}$
	2^{19}	2^5	$2^{4.76}$
	2^{20}	2^6	$2^{6.56}$
	2^{21}	2^7	$2^{6.10}$
	2^{22}	2^8	$2^{7.80}$
	2^{23}	2^9	$2^{9.12}$
	2^{24}	2^{10}	$2^{9.51}$
	2^{25}	2^{11}	$2^{10.06}$

Figure 5 provides a visual comparison between theoretical expectations and empirical results from Table 6 for 2-round and 3-round differential distinguishers. The x-axis represents the number of plaintext pairs tested (N), while the y-axis shows the number of successful differential matches.

4 Conclusion

In this paper, we have analyzed the block cipher designed in¹⁷ against differential cryptanalysis. The cipher works on 64-bit blocks with a 64-bit key. Its design uses modular addition, the XOR operator, and rotation.

We have shown that reduced-round differential distinguishers can be constructed for this cipher. We experimentally verified the probability of the obtained differential distinguishers for up to three rounds. Based on our results, the full-round version of this cipher (five rounds) can be distinguished from a random 64-bit permutation using approximately 2^{41} chosen plaintext. Therefore, the designers' claim about the cipher's security is not reliable.

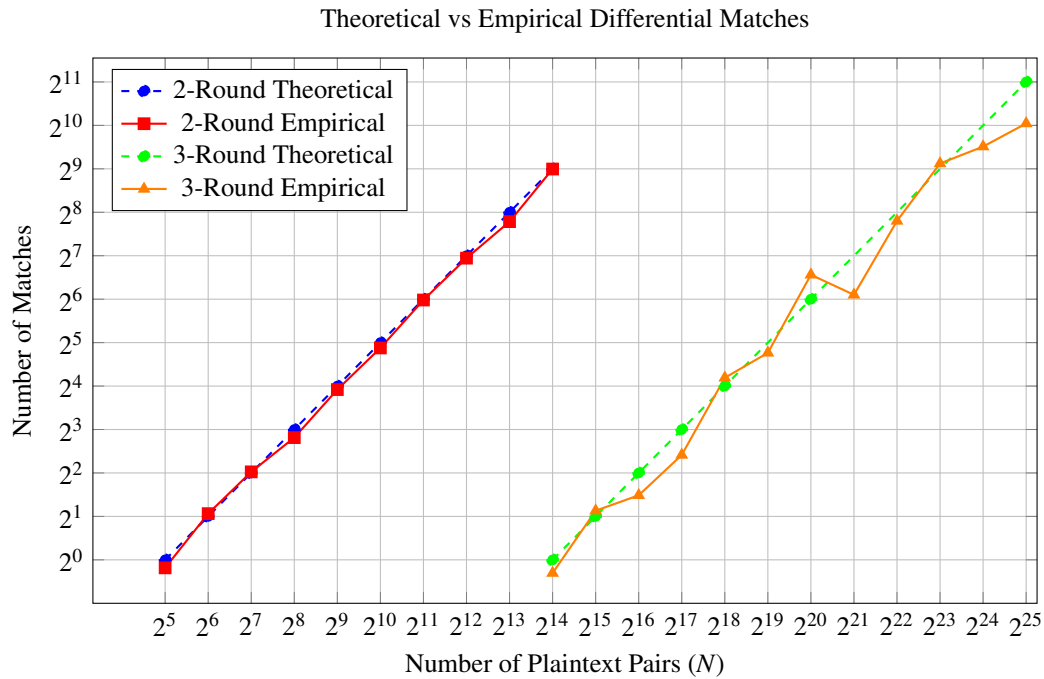


Figure 5. Comparison of theoretical and empirical differential matches for 2-round and 3-round distinguishers with theoretical probabilities 2^{-5} and 2^{-14} , respectively.

As many cipher algorithms are being developed today for various purposes (such as IoT, image encryption, watermarking, etc.), paying attention to known attacks is crucial. Designers often aim to keep their designs as lightweight and straightforward as possible to improve efficiency. However, resistance to various attacks must also be carefully considered during the design process. In this paper, we aim to demonstrate the importance of this issue by analyzing the target block cipher.

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